

TAYLOR PENTZ

Senior Software Engineer

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Professional Summary

Experienced and innovative Senior ML Engineer with 8+ years in software development and 4+ years in deploying vision-based machine learning models to production. Passionate about contributing to cutting-edge generative AI systems and building scalable, multimodal solutions. Adept at collaborating across research and engineering teams to bring models from ideation to real-world impact.

Technical Skills

- ❖ **Languages & ML:** Python, PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, FastAPI
- ❖ **Cloud:** AWS (S3, EC2, SageMaker, Lambda), GCP (Vertex AI, BigQuery, Cloud Run), Azure
- ❖ **Databases:** PostgreSQL, MongoDB, Redis, MySQL
- ❖ **Tools:** Docker, Kubernetes, Git, DVC, MLflow, Airflow
- ❖ **MLOps & DevOps:** CI/CD, Terraform, Prometheus, Grafana, Weights & Biases
- ❖ **Vision Models:** CNNs, ViT, YOLOv5, CLIP, VQA, multimodal fusion networks
- ❖ **Other:** REST APIs, GraphQL, Kafka, NoSQL, Linux, Agile, Scrum

Professional Experience

SumatoSoft

Senior Software Engineer
Boston, MA (Remote) | Apr 2023 – Mar 2025

- Designed and deployed scalable vision-language models using CLIP and ViT architecture, enhancing search and classification accuracy by 32% across multimodal datasets.
- Built end-to-end ML pipelines for training, evaluation, and deployment of models on AWS SageMaker, integrating DVC and MLflow for model versioning and tracking.
- Collaborated with product and research teams to productionize models, improving iteration velocity and reducing deployment time by 40%.
- Deployed real-time inference systems using FastAPI and Docker, serving multimodal models with sub-second latency for client applications.
- Led the integration of Weights & Biases for experiment tracking across model development lifecycle.
- Optimized preprocessing pipelines using OpenCV and TensorRT to reduce data loading and inference times by 25%.
- Created custom MLOps workflows in Kubernetes for continuous training and automated retraining triggered by data drift detection.
- Mentored junior ML engineers and facilitated regular knowledge-sharing sessions on vision models and MLOps practices.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 – Mar 2023

- Developed Python-based microservices supporting real-time image and video data pipelines for surveillance and logistics applications.
- Contributed to the design of a hybrid model combining CNN and LSTM for visual context-aware object tracking in warehouses.
- Implemented CI/CD workflows for machine learning systems using GitHub Actions, Docker, and Kubernetes.
- Worked closely with computer vision researchers to convert Jupyter-based experiments into scalable backend services.
- Integrated Redis for fast in-memory data access during high-throughput model inference operations.
- Collaborated with product teams to improve UX in labeling interfaces, reducing annotation turnaround time by 18%.
- Utilized GCP Cloud Run and BigQuery to deploy and analyze the behavior of deployed models across thousands of edge devices.
- Maintained code quality with test coverage above 85% using Pytest and Black for formatting.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 – Jun 2018

- Engineered robust backends for eCommerce and health platforms, leveraging Python, Flask, and PostgreSQL.
- Built RESTful APIs for customer data aggregation pipelines integrated with ML-driven personalization features.
- Integrated Apache Kafka for event-driven user activity tracking and model feedback loops.
- Partnered with external ML consultants to operationalize anomaly detection algorithms.
- Managed PostgreSQL and NoSQL schema evolution as new data types were introduced into the ML workflows.
- Automated ETL pipelines to transform raw user behavior data into clean datasets ready for model training.
- Containerized ML workloads using Docker Compose and deployed via Azure DevOps pipelines.
- Mentored junior developers transitioning into ML and data engineering roles.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 – Feb 2015

- Contributed to an early prototype of a medical image diagnosis tool, training initial CNN-based models for X-ray classification.
- Built data labeling tools in React for radiologist workflow, improving annotation throughput.
- Developed internal scripts in Python for preprocessing and augmentation of radiology images.
- Created REST APIs for fetching predictions and logs from early inference endpoints.
- Assisted in migrating on-prem workloads to AWS EC2 and integrating with S3 buckets.
- Supported efforts to integrate model feedback loops into reporting dashboards using SQL + NoSQL databases.

- Deployed automated model evaluation notebooks using Jupyter + Papermill on scheduled intervals.
- Participated in weekly research sprint reviews and demoed incremental results.

Projects

- ✓ **Multimodal Search Engine (2024)**
Developed a multimodal search engine using vision-language models (CLIP) for a retail recommendation platform. Achieved 34% higher engagement through semantic search on image + text pairs.
- ✓ **Real-Time Incident Detection (2022)**
Implemented a real-time ML-powered incident detection system in a logistics fleet, combining object tracking and NLP-based log analysis. Integrated via GCP Pub/Sub and Cloud Run.
- ✓ **Medical Image Classifier (2016-2017)**
Contributed to a medical image classification MVP with CNN models for disease screening, used as proof-of-concept in national health grant.

Certifications

- ❖ AWS Certified Machine Learning - Specialty
- ❖ Google Cloud Professional Machine Learning Engineer
- ❖ TensorFlow Developer Certificate

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science (2010 – 2014)